

Computer Output and Standard Deviation of Residuals

Sometimes called “MiniTab”, computer output on the AP exam is a quick way to give statistical information on linear regression. The key is to remember what all the information standards for.

Predictor	Coef	St Dev	t ratio	P
Constant	3.371	1.337	2.52	.065
Age	2.1143	0.2321	9.11	.001

$s = 0.9710$	$R\text{-sq} = 95.4\%$	$R\text{-sq(adj)} = 94.3\%$
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These values will be used in Unit 9. Ignore them for now.

Always ignore this value. We do not use it in AP Statistics.

Using these four pieces of information, we can also get the following:

LSRL: _____ $r =$ _____

The standard deviation of the residuals (“s”) will be used more in Unit 9, but we can interpret it here. This value gives the approximate size of a “typical” prediction error (residual). Larger values of “s” means our line is expected to give larger residuals. The units for the standard deviation of residuals is the same of the residuals (and of the y-values). So it depends on the data for what is considered a “large” residual error.

For example, if “s” was 100, that would be a large value if our y-value was trying to predict age in years but it would be a small residual if we were trying to predict age in seconds. Context is important when interpreting this value.

Influential Points

An influential point in a data set, is a point that has leverage on the correlation and regression line. In other words, when removed, this point changes the regression line substantially. An influential point might be considered an outlier if it does not fit with the overall pattern of the data, but an influential point might also fit the data, but change the regression line and/or correlation significantly when removed.

Example - Here is some hypothetical data:							Draw a scatterplot.
X	1	2	3	4	5	6	
Y	1	5	3	5	4	8	

LSRL: _____ r: _____

→ Remove the ordered pair (2, 5). The point is not considered an outlier because it fits the rest of the data, but we will examine the impact on the LSRL and correlation to see that it is influential.

LSRL: _____ r: _____

a) What happened to the slope? Why?

b) What happened to the y-intercept? Why?

c) What happened to the correlation? Why?

Departures from Linearity

The AP Statistics course deals only with two-variable data that can be modeled by a line OR nonlinear two-variable data that can be transformed in such a way that the transformed data can be modeled by a line.

To know what transformation to use, graph the data first. If the scatterplot does not show a linear pattern, or the residual plot pattern is not random, consider a transformation.

You can transform the independent variable, dependent variable, or both. There are a wide range of transformations you can do, but the AP exam will keep it simple. Here is an example of one of the most common transformations you will encounter.

Example: Natural Log Transformation

The number of a certain type of bacteria present (in thousands) after a certain number of hours is given in the following table.

Hours	Number	Create a scatterplot on your calculator and sketch it here.	Create a residual plot on your calculator and sketch it here.
1.0	1.8		
1.5	2.4		
2.0	3.1		
2.5	4.3		
3.0	5.8		
3.5	8.0		
4.0	10.6		
4.5	14.0		
5.0	18.0		

Why is a linear model not appropriate here?

- What parent function does this scatterplot look like? Exponential!
- Think: How do we “undo” exponents in an equation? _____
- For this problem (you can do some trial and error if you want, or you can trust me) the natural logarithm will be the best way to transform the data to achieve linearity.
- On the AP exam, to save time, if you needed to do this by hand, they would tell you the transformation.

			Create a scatterplot (between Hours and LN(Number)) on your calculator and sketch it here.	Create a residual plot (between Hours and LN(Number)) on your calculator and sketch it here.
Hours	Number	LN(Number)		
1.0	1.8	0.59		
1.5	2.4	0.88		
2.0	3.1	1.13		
2.5	4.3	1.46		
3.0	5.8	1.76		
3.5	8.0	2.08		
4.0	10.6	2.36		
4.5	14.0	2.64		
5.0	18.0	2.89		

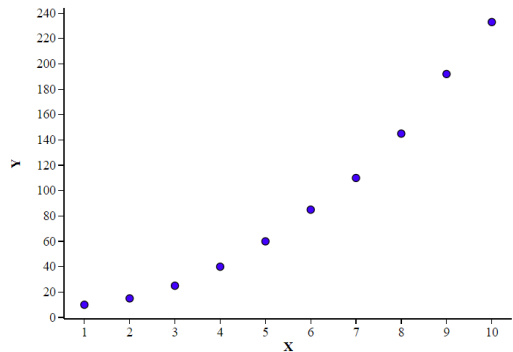
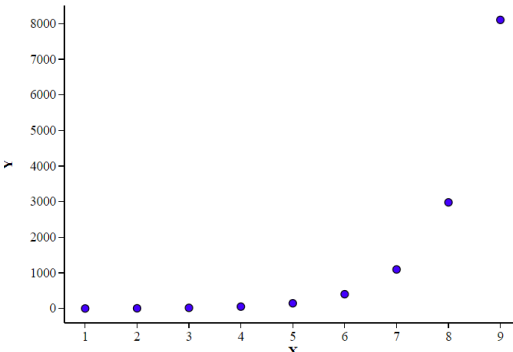
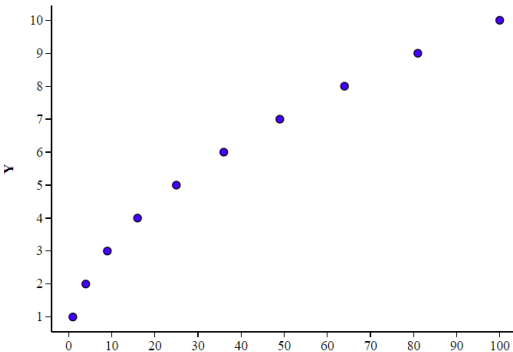
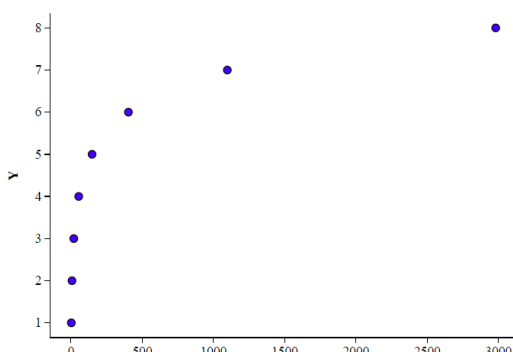
Why is this a better model for linear regression?

The regression equation for the transformed data is _____

Using the equation above, what would be the predicted quantity of bacteria after 3.75 hours?

There are many types of transformations available and you can even get really crafty and add in some constants!

For AP Statistics, however, we keep it simple. Here are some common patterns that you might see and need to transformation:

Intuitive Line of Best Fit	Suggested Transformation	Example Scatterplot
Contains (0, 0) and appears to have a slight curve.	$(x, y) \rightarrow (\ln x, \ln y)$	
Contains a non-zero y-intercept and appears exponential.	$(x, y) \rightarrow (x, \ln y)$	
Contains (0,0) and appears logarithmic.	$(x, y) \rightarrow (\sqrt{x}, y)$	
Contains a non-zero y-intercept and appears logarithmic.	$(x, y) \rightarrow (\ln x, y)$	

To test and see which model would be appropriate, you would compare the scatterplots, r and r^2 , and the residual plots of both the original and transformed data.